

```
62. # Compile results
63. metrics = {
64.     'Total Return': total_return,
65.     'Annualized Return': annualized_return,
66.     'Volatility': volatility,
67.     'Sharpe Ratio': sharpe_ratio,
68.     'Maximum Drawdown': max_drawdown,
69.     'Calmar Ratio': calmar_ratio,
70.     'Win Rate': win_rate,
71.     'Average Profit': avg_profit,
72.     'Average Loss': avg_loss,
73.     'Profit/Loss Ratio': profit_loss_ratio,
74.     'Total Trades': len(trades),
75.     'Trading Costs': backtest_results['Trade Cost'].sum()
76. }
77.
78. return metrics
79.
80.
```

This evaluation function calculates a comprehensive set of metrics that help assess not just profitability, but also risk, consistency, and efficiency—factors that are critical for long-term trading success.

Parameter Optimization Techniques

Every trading strategy has parameters that can be adjusted—whether it's moving average periods, RSI thresholds, or position sizing rules. Optimization is the process of finding the parameter values that produce the best results. But there's a critical balance to strike: optimize too little, and you leave performance on the table; optimize too much, and you risk creating a strategy that works beautifully in backtests but fails miserably in real trading.

Let's explore common optimization techniques and how to use them wisely.

Grid Search: The Systematic Approach

Grid search is perhaps the simplest optimization technique: you define a range of values for each parameter, test all possible combinations, and identify the best-performing set.